

9.23 For TE mode, $\vec{E}_{lmn}^{(M)} = Z_0 j_\ell(kr) \vec{L} Y_{lm}(\theta, \phi)$, and

$$\vec{H}_{lmn}^{(M)} = \vec{n} \frac{\ell(\ell+1)}{kr} j_\ell(kr) Y_{lm}(\theta, \phi) - \frac{i}{kr} \frac{\partial}{\partial r} [r j_\ell(kr)] \vec{n} \times \vec{L} Y_{lm}(\theta, \phi).$$

With $k = \chi_{lm}/a$. By Eq. (8.58), the power loss is

$$\begin{aligned} P_{\text{loss}} &= \frac{1}{2\sigma\delta} \int |\vec{n} \times \vec{H}|^2 ds = \frac{1}{2\sigma\delta} \frac{1}{k^2} \left\{ \frac{\partial}{\partial r} [r j_\ell(kr)] \right\}_{r=a}^2 \int |\vec{L} Y_{lm}(\theta, \phi)|^2 d\Omega \\ &= \frac{1}{2\sigma\delta} \frac{\ell(\ell+1)}{k^2} \left\{ \frac{\partial}{\partial r} [r j_\ell(kr)] \right\}_{r=a}^2. \end{aligned}$$

The total energy in the cavity is

$$U = \frac{\epsilon_0}{2} \int |\vec{E}|^2 d^3x = \frac{\mu_0}{2} \int_0^a r^2 j_\ell(kr)^2 dr \int |\vec{L} Y_{lm}|^2 d\Omega = \frac{\mu_0}{2} \ell(\ell+1) \int_0^a r^2 j_\ell(kr)^2 dr$$

We will show how to perform the integral here. By the definition of spherical Bessel function,

$j_\ell(r) = \sqrt{\frac{\pi}{2r}} J_{\ell+1/2}(r)$, we have

$$\int_0^a r^2 j_\ell(kr)^2 dr = \frac{\pi}{2k} \int_0^a r J_{\ell+1/2}(\chi_{lm} \frac{r}{a})^2 dr = \frac{\pi}{2k} \cdot \frac{a^2}{2} J_{\ell+3/2}(ka)^2 = \frac{a^3}{2} j_{\ell+1}(ka)^2.$$

Also, $\frac{\partial}{\partial r} [r j_\ell(kr)] \Big|_{r=a} = j_\ell(ka) + ka j_\ell'(ka) = ka j_{\ell+1}(ka)$, then

$$Q = \omega \frac{U}{P_{\text{loss}}} = \omega \frac{\frac{\mu_0}{2} \ell(\ell+1) \frac{a^3}{2} j_{\ell+1}(ka)^2}{\frac{1}{2\sigma\delta} \frac{\ell(\ell+1)}{k^2} k^2 a^2 j_{\ell+1}(ka)^2} = \omega \mu_0 \sigma \delta \cdot \frac{a}{2}.$$

But $\delta = (2/\mu_0 \omega \sigma)^{1/2}$, then

$$Q = a \sqrt{\frac{\mu_0 \omega \sigma}{2}} = \frac{a}{\delta}, \quad \text{for all TE modes.}$$

For TM mode, $\vec{H}_{lmn}^{(E)} = j_\ell(k'r) \vec{L} Y_{lm}(\theta, \phi)$, and

$$\vec{E}_{lmn}^{(E)} = -\vec{n} \frac{Z_0 \ell(\ell+1)}{k'r} j_\ell(k'r) Y_{lm} + \frac{i Z_0}{k'r} \frac{\partial}{\partial r} [r j_\ell(k'r)] \vec{n} \times \vec{L} Y_{lm}.$$

Similarly, the power loss is

$$P_{\text{loss}} = \frac{1}{2\sigma\delta} \int |\vec{n} \times \vec{H}_{\text{in}}|^2 dS = \frac{a^2}{2\sigma\delta} \mu(\ell+1) j_\ell(k'a)^2, \text{ and calculate energy from magnetic field}$$

$$U = \frac{\mu_0}{2} \int |\vec{H}|^2 d^3x = \frac{\mu_0}{2} \int_0^a r^2 j_\ell(k'r)^2 dr \int |\vec{L} Y_{\ell m}|^2 d\Omega = \frac{\mu_0}{2} \mu(\ell+1) \int_0^a r^2 j_\ell(k'r)^2 dr$$

Using the integration identity

$$\int_0^a r^2 j_\ell(k'r)^2 dr = \frac{a^3}{2} \left(j_\ell(k'a)^2 - j_{\ell-1}(k'a) j_{\ell+1}(k'a) \right) + C,$$

we have
$$U = \frac{\mu_0}{2} \mu(\ell+1) \frac{a^3}{2} \left(j_\ell(k'a)^2 - j_{\ell-1}(k'a) j_{\ell+1}(k'a) \right)$$

To simplify the energy expression, from the recursion relations Eq. (9.90), we have

$$\frac{d}{dx} [x j_\ell(x)] = x j_{\ell-1}(x) - \ell j_\ell(x). \quad \text{Set } x = k'a, \text{ by the equations for characteristic frequencies,}$$

the left hand side is 0, and thus, $j_{\ell-1}(k'a) = \frac{\ell}{k'a} j_\ell(k'a)$. Then, using

$$\frac{\ell+1}{x} j_\ell(x) = j_{\ell+1}(x) + j_{\ell-1}(x), \text{ we have } j_{\ell+1}(k'a) = \frac{\ell+1}{k'a} j_\ell(k'a) - \frac{\ell}{k'a} j_\ell(k'a) = \frac{\ell+1}{k'a} j_\ell(k'a)$$

Therefore,
$$U = \frac{\mu_0}{2} \mu(\ell+1) \frac{a^3}{2} j_\ell(k'a)^2 \left(1 - \frac{\ell(\ell+1)}{(k'a)^2} \right), \text{ and}$$

$$Q = \omega \frac{U}{P_{\text{loss}}} = \omega \cdot \frac{\frac{\mu_0 a}{2}}{\frac{1}{\sigma\delta}} \left(1 - \frac{\ell(\ell+1)}{(k'a)^2} \right) = \frac{a}{\delta} \left(1 - \frac{\ell(\ell+1)}{\chi_{\ell n}'^2} \right), \text{ where } \chi_{\ell n}' \text{ is the } n\text{th solution}$$

to the equation
$$\frac{d}{dx} [x j_\ell(x)] = 0$$